

Web Engineering & Development (SWE 363)

Back-end Development Fundamentals

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In today's lecture:

- Express.js
 - Web Servers
 - Routes
 - Middleware

Reference:

- Zybook: 6.4

6.4 Express.js

What is Express.js?

| **Express.js** is a web framework for Node.js

Express simplifies:

- Creating web servers
- Handling HTTP requests and responses
- Defining routes (URLs)
- Using middleware

Think of Express as a toolkit that makes building web servers much easier!

Why Use Express.js?

Without Express:

- Write lots of code
- Handle HTTP manually
- Complex routing logic
- More time-consuming

With Express:

- Simple, clean code
- Built-in HTTP handling
- Easy route definition
- Fast development

Installing Express

Create a new project:

```
npm init -y
```

Install Express:

```
npm install express
```

Install Cross-Origin Resource Sharing (CORS) (for frontend communication):

```
npm install cors
```

Creating Your First Express Server

Basic server setup:

```
import express from "express";

const app = express();
const PORT = 3000;

app.listen(PORT, () => {
  console.log(`Server running on http://localhost:${PORT}`);
});
```

Run the server:

```
node server.js
```

What is a Web Server?

| A **web server** listens for requests and sends responses

Simple analogy:

- **Restaurant:** Server listens for orders (requests)
- **Kitchen:** Server processes the order
- **Waiter:** Server brings food back (response)

In web development:

- Browser sends a **request** (asks for data)
- Server processes the request
- Server sends a **response** (returns data)

Understanding Requests and Responses

Request (req)

- Data **from** the client
- URL, method (GET/POST)
- Headers, body
- Query parameters

Response (res)

- Data **to** the client
- Status code (200, 404, etc.)
- JSON, HTML, text
- Headers

What are Routes?

Routes define what happens when someone visits a specific URL

Example:

- `/` → Home page
- `/api/quote` → Get a random quote
- `/api/users` → Get user data

Routes organize your API - each URL does something different!

Why Do We Need Routes?

- **Organization:** Each URL has a specific purpose
- **Clarity:** Easy to understand what each endpoint does
- **Separation:** Different features in different routes
- **Scalability:** Easy to add new features

Example:

```
/api/students → Student data  
/api/teachers → Teacher data  
/api/courses → Course data
```

Creating Your First Route

Basic GET route:

```
app.get("/", (req, res) => {  
  res.send("Welcome to my API!");  
});
```

What this does:

- When someone visits `http://localhost:3000/`
- Server responds with: "Welcome to my API!"

Route Syntax

General format:

```
app.METHOD(path, handler)
```

Components:

- `app` → Express application
- `METHOD` → HTTP method (get, post, put, delete)
- `path` → URL path (e.g., `/api/quote`)
- `handler` → Function that handles the request

HTTP Methods

Method	Purpose	Example
GET	Retrieve data	Get a quote
POST	Create data	Create a user
PUT	Update data	Update user info
DELETE	Delete data	Delete a user

GET Route Example

Simple GET route:

```
app.get("/api/quote", (req, res) => {  
  const quote = "Talk is cheap. Show me the code.";  
  res.json({ quote: quote });  
});
```

Test it:

- Visit: `http://localhost:3000/api/quote`
- Response: `{ "quote": "Talk is cheap. Show me the code." }`

Sending Different Response Types

Send plain text:

```
res.send("Hello World");
```

Send JSON:

```
res.json({ message: "Success", data: [...] });
```

Send status code:

```
res.status(404).send("Not Found");
```


Multiple Routes Example

```
app.get("/", (req, res) => {  
  res.send("Welcome to the Quote Generator API");  
});  
  
app.get("/api/quote", (req, res) => {  
  const quote = getRandomQuote();  
  res.json({ quote });  
});  
  
app.get("/api/about", (req, res) => {  
  res.json({  
    name: "Quote API",  
    version: "1.0.0"  
  });  
});
```

What is Middleware?

Middleware are functions that run between a request and a response

Think of middleware as:

- **Security guards** checking requests
- **Helpers** that process data before it reaches your route
- **Loggers** that record what's happening

Middleware Flow

Request → Middleware 1 → Middleware 2 → Route Handler → Response

Example:

1. Request arrives
2. CORS middleware checks origin
3. Logger middleware records request
4. Route handler processes request
5. Response sent back

Using Middleware

Basic middleware:

```
app.use((req, res, next) => {  
  console.log("Request received:", req.url);  
  next(); // Continue to next middleware/route  
});
```

Important: Always call `next()` to continue!

CORS Middleware

CORS (Cross-Origin Resource Sharing) allows frontend to communicate with backend

Why we need it:

- Frontend runs on `http://localhost:5173`
- Backend runs on `http://localhost:3000`
- Browsers block cross-origin requests by default

Solution:

```
import cors from "cors";  
app.use(cors());
```

CORS Example

Without CORS:

Frontend (port 5173) → ❌ Blocked → Backend (port 3000)

With CORS:

```
import cors from "cors";  
app.use(cors());
```

Frontend (port 5173) → ✅ Allowed → Backend (port 3000)

Built-in Express Middleware

Parse JSON data:

```
app.use(express.json());
```

Parse URL-encoded data:

```
app.use(express.urlencoded({ extended: true }));
```

These allow your server to read request body data!

Complete Server Example

```
import express from "express";
import cors from "cors";

const app = express();
const PORT = 3000;

// Middleware
app.use(cors());
app.use(express.json());

// Routes
app.get("/", (req, res) => {
  res.send("Welcome to the Quote Generator API");
});
app.listen(PORT, () => {
  console.log(`Server running on http://localhost:${PORT}`);
});
```


Request Object (req)

Common properties:

```
app.get("/api/user/:id", (req, res) => {  
  req.params.id      // URL parameters  
  req.query.search   // Query string (?search=hello)  
  req.body           // Request body (POST data)  
  req.headers        // Request headers  
  req.url            // Request URL  
});
```

Response Object (res)

Common methods:

```
res.send("Text")           // Send text
res.json({ data })        // Send JSON
res.status(404)            // Set status code
res.sendStatus(200)        // Send status + message
res.redirect("/")          // Redirect to URL
```

Route Parameters

Dynamic routes:

```
app.get("/api/user/:id", (req, res) => {  
  const userId = req.params.id;  
  res.json({ userId });  
});
```

Example:

- URL: `/api/user/123`
- `req.params.id` = `"123"`

Query Parameters

Query strings:

```
app.get("/api/search", (req, res) => {  
  const query = req.query.q;  
  res.json({ search: query });  
});
```

Example:

- URL: `/api/search?q=javascript`
- `req.query.q` = `"javascript"`

POST Route Example

Handle POST requests:

```
app.post("/api/users", (req, res) => {  
  const newUser = req.body;  
  // Save user to database  
  res.status(201).json({  
    message: "User created",  
    user: newUser  
  });  
});
```

Note: Requires `app.use(express.json())` middleware!

Error Handling

Basic error handling:

```
app.get("/api/data", (req, res) => {  
  try {  
    // Your code here  
    res.json({ data: "success" });  
  } catch (error) {  
    res.status(500).json({  
      error: "Something went wrong"  
    });  
  }  
});
```

package.json Review

What is package.json?

- Configuration file for your project
- Lists all dependencies
- Defines project metadata
- Contains scripts

```
{  
  "name": "quote-api",  
  "version": "1.0.0",  
  "dependencies": {  
    "express": "^4.18.2",  
    "cors": "^2.8.5" }  
}
```

package-lock.json

What is package-lock.json?

- Records **exact versions** of all packages
- Ensures **consistent installs**
- **Auto-generated** - don't edit manually!

Why it matters:

- Team members get same versions
- Production deployments are predictable
- Prevents "works on my machine" issues

Summary Table

Concept	Purpose	Example
Express	Web framework	<code>const app = express()</code>
Route	Handle URL	<code>app.get("/", handler)</code>
Middleware	Process requests	<code>app.use(cors())</code>
req	Request data	<code>req.params , req.body</code>
res	Send response	<code>res.json() , res.send()</code>

30m

Demo

Starter code at: <https://classroom.github.com/a/RFGnUwIL>

- Set up Express server
- Create routes
- Use middleware (CORS)
- Connect frontend to backend